

Parallel & Distributed Computing



SYMMETRIC AND ASYMMETRIC MULTIPROCESSING



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Lesson from Quran

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Lecture Outline

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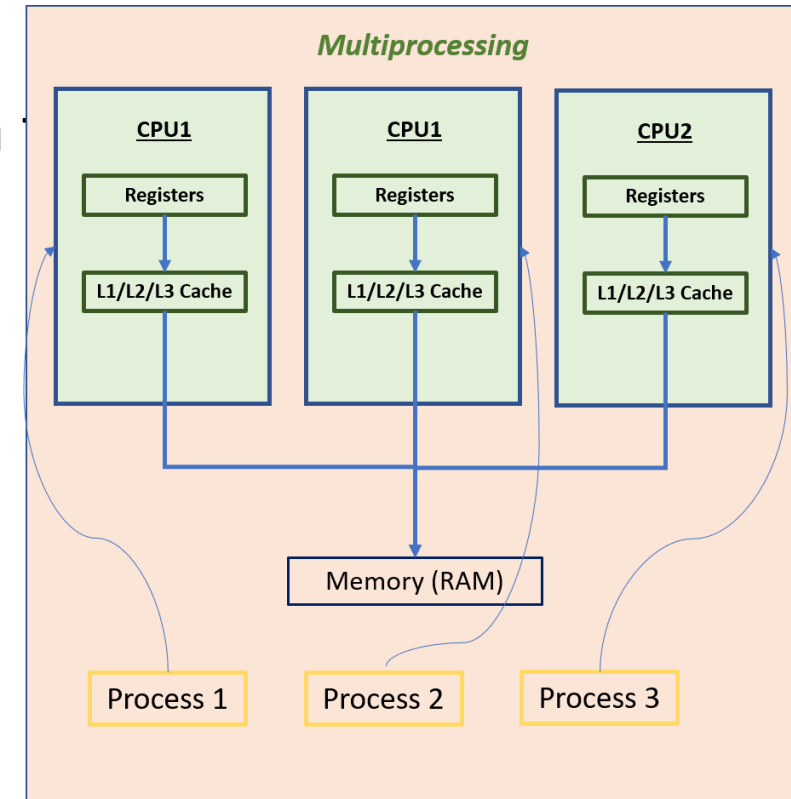
- Introduction to Multiprocessing
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- SMP Characteristics, Advantages & Disadvantages
- Asymmetric Multiprocessing (AMP)
- AMP Characteristics, Advantages & Disadvantages
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Introduction to Multiprocessing

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- ❑ Multiprocessing is a method of using **two or more processors (CPUs)** within a single computer system execute multiple processes simultaneously.
- ❑ To increase the overall **system throughput** and **performance**.
- ❑ **Characteristics:**
 - ▣ Multiple CPUs share the same main memory and I/O.
 - ▣ Tasks are divided and executed in parallel.
 - ▣ Operating system coordinates resource allocation.
- ❑ **Common in:**
Servers, data centers, HPC systems, and modern multi-core processors.



Why Multiprocessing?

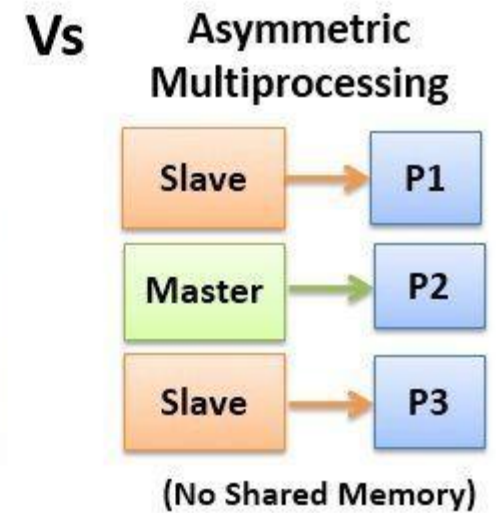
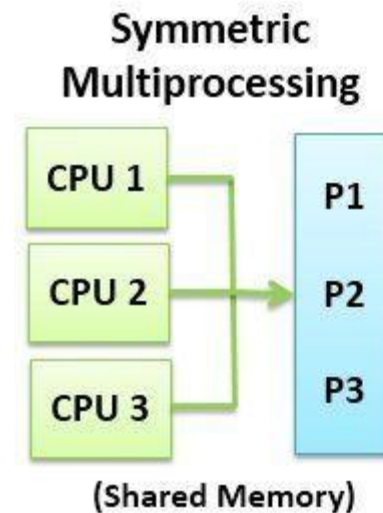
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- **Performance:** Speed up computation by dividing tasks among multiple CPUs.
- **Reliability:** Even if one CPU fails, others can continue executing.
- **Scalability:** Adding CPUs increases throughput up to a certain limit.
- **Parallelism:** Enables true parallel execution, unlike time-sharing.
- **Examples:**
 - ▣ Scientific simulations
 - ▣ Databases
 - ▣ Cloud servers
 - ▣ Embedded real-time controllers

Types of Multiprocessing Systems

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- ❑ Multiprocessing systems are categorized mainly as:
 - ❑ **Symmetric Multiprocessing (SMP):**
All processors are **identical** and share a **common memory** and I/O.
 - ❑ **Asymmetric Multiprocessing (AMP):**
Processors have **different roles**—one acts as the **master**, others as **slaves**.
 - ❑ The key difference lies in **task scheduling, resource control, and OS management**.



Symmetric Multiprocessing (SMP)

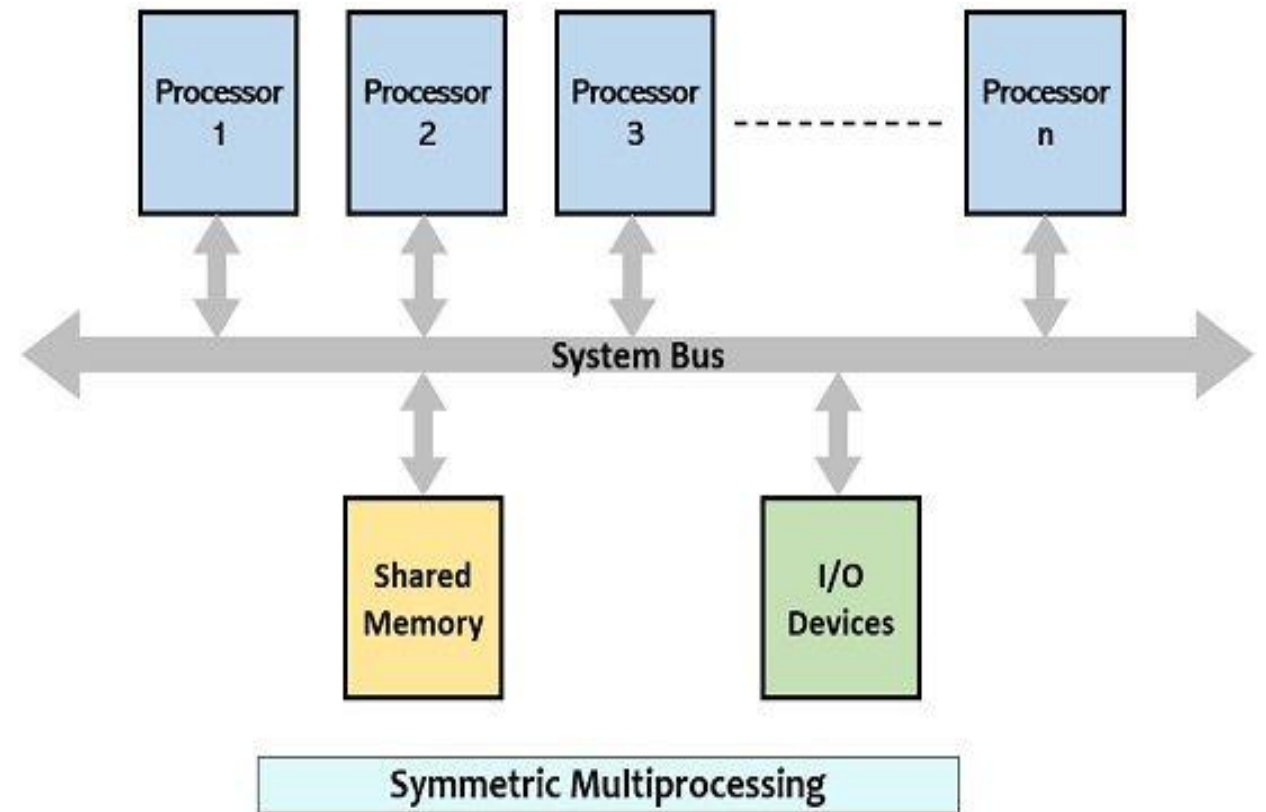
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- ❑ In SMP, **each processor is equal** in terms of capability and responsibility.
- ❑ All processors share:
 - ❑ **Main memory**
 - ❑ **I/O subsystem**
 - ❑ **Operating system** (single OS image)
- ❑ **Each processor runs a copy of the OS**, and the OS dynamically schedules tasks to any processor.
- ❑ SMP is the foundation of **modern multi-core architectures**.

SMP System Architecture

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- ❑ Multiple identical CPUs connected to shared memory through a **system bus** or **crossbar switch**.
- ❑ Each CPU has:
 - ❑ Its own **cache** for fast local access.
 - ❑ Equal access rights to the **shared memory** and **I/O**.
- ❑ **Inter-processor communication** occurs through shared memory.



SMP Characteristics

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- ❑ **Homogeneous processors:** All are identical in function and capability.
- ❑ **Shared memory model:** All CPUs access a single main memory.
- ❑ **Single OS:** Manages all resources collectively.
- ❑ **Uniform memory access (UMA):** Each CPU has equal latency to memory.
- ❑ **Dynamic scheduling:** The OS assigns any process/thread to any CPU.
- ❑ **Communication:** Done via shared memory instead of message passing.

Operating System Role in SMP

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- ❑ **Process scheduling:** Balances workload among CPUs (load balancing).
- ❑ **Synchronization:** Manages access to shared data to avoid race conditions.
- ❑ **Memory management:** Ensures consistent access to memory for all CPUs.
- ❑ **Interrupt handling:** Any CPU can handle interrupts.
- ❑ **Example OS:** Linux, Windows Server, macOS — all are SMP-capable.

SMP Advantages and Disadvantages

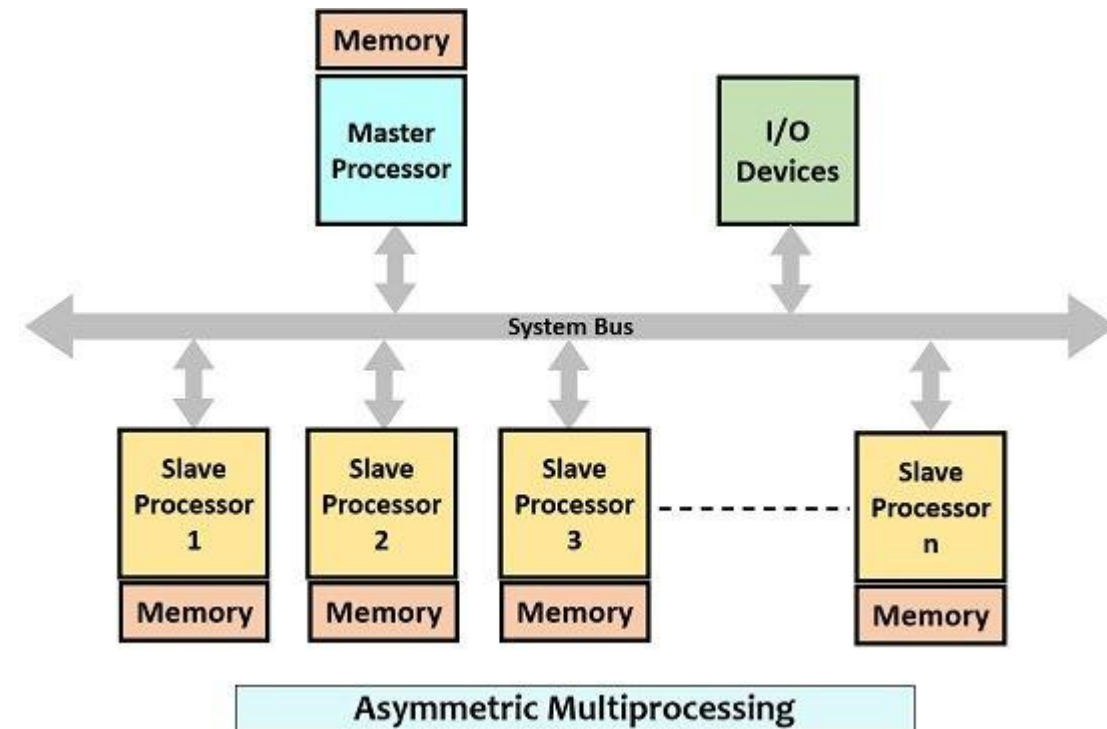
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- **Improved Performance:**
Multiple processors execute tasks in parallel.
- **Reliability and Fault Tolerance:**
One CPU failure doesn't halt the system.
- **Resource Sharing:**
CPUs share memory and devices efficiently.
- **Load Balancing:**
The OS evenly distributes tasks.
- **Scalability:**
More processors can be added (up to hardware limits).
- **SMP Bus Contention:**
Shared bus becomes a bottleneck when CPUs increase.
- **Cache Coherence Problem:**
Each CPU's cache may hold outdated data → requires coherence protocols.
- **Memory Bottleneck:**
All CPUs compete for access to main memory.
- **Limited Scalability:**
Performance gain decreases as more CPUs are added.
- **Complex OS Design:**
Requires advanced synchronization and scheduling.

Asymmetric Multiprocessing (AMP)

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- ❑ In AMP, processors have **different roles and responsibilities**.
- ❑ **Master CPU:**
 - ▣ Controls system operation and manages I/O, memory, and task distribution.
 - ▣ Runs the **main operating system**.
- ❑ **Slave CPUs:**
 - ▣ Perform **specific assigned tasks** only.
 - ▣ Operate under the control of the master CPU.
- ❑ Common in **embedded systems, robotics, and IoT**.



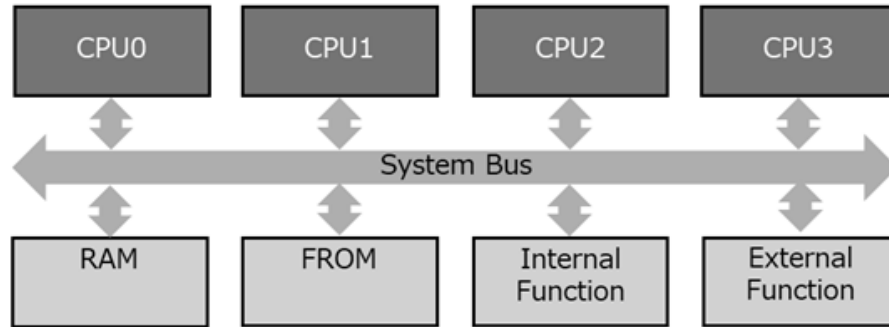
AMP System Architecture

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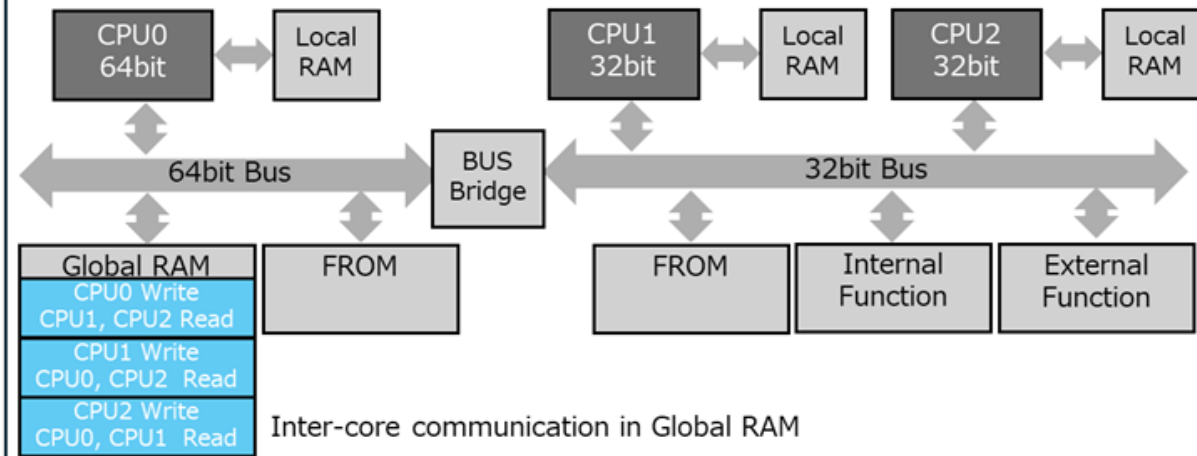


SMP and AMP

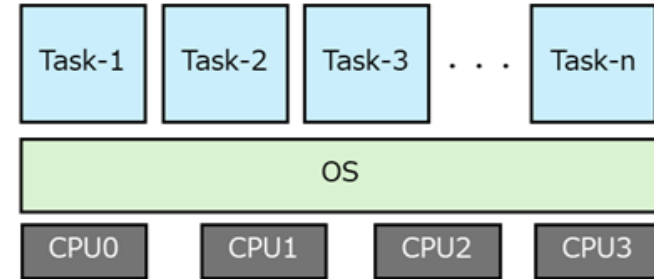
SMP (Symmetric Multi-Processing)



AMP (Asymmetric Multi-Processing)

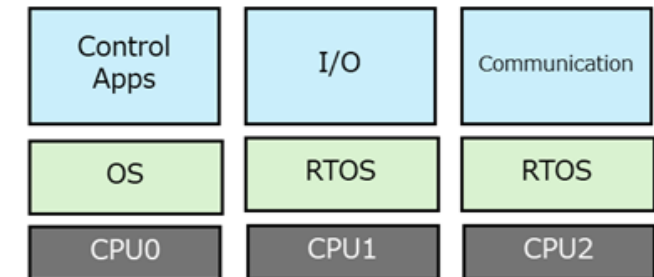


SMP processing



Load balancing type
OS scheduler allocates tasks equally to each processor, improving performance.

AMP Processing



Functionally distributed type
Same configurations as single core, but each core has a different function.

AMP Characteristics

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- ❑ **Master-slave relationship:** One master controls all operations.
- ❑ **Single OS:** Only the master runs the OS; slaves run firmware or RTOS.
- ❑ **Static task assignment:** Each processor has predefined functions.
- ❑ **Communication:** Through shared memory or message queues.
- ❑ **Less interdependence:** Slaves are isolated from each other.
- ❑ **Typical usage:** Systems requiring determinism and real-time response.

AMP Advantages and Disadvantages

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- **Simplified Design:**
Hardware and control logic are simpler.
- **Efficient for Dedicated Tasks:**
Slaves specialize in certain functions, improving efficiency.
- **Real-time Performance:**
Predictable timing due to dedicated task allocation.
- **Lower Overhead:**
Less OS overhead because only one processor manages resources.
- **Master Dependency:**
Master failure → system halt.
- **Poor Scalability:**
Adding new processors requires redesign of control logic.
- **Inefficient Utilization:**
Slaves remain idle when their assigned tasks are finished.
- **Complex Interprocessor Communication:**
Requires explicit synchronization mechanisms.
- **Limited Flexibility:**
Tasks cannot be dynamically reassigned.

Comparison Table – SMP vs AMP

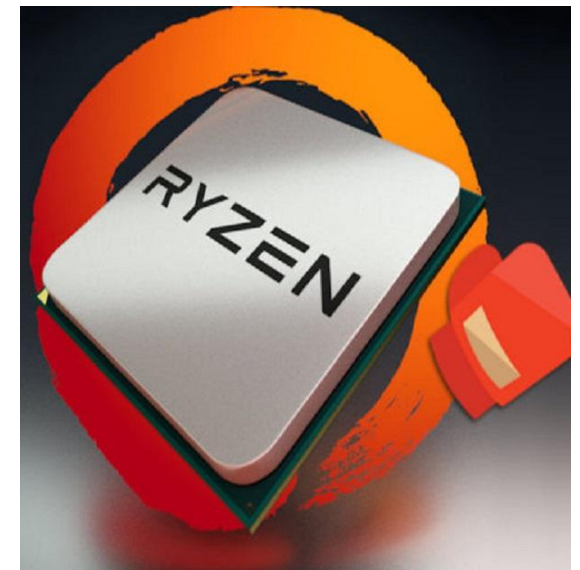
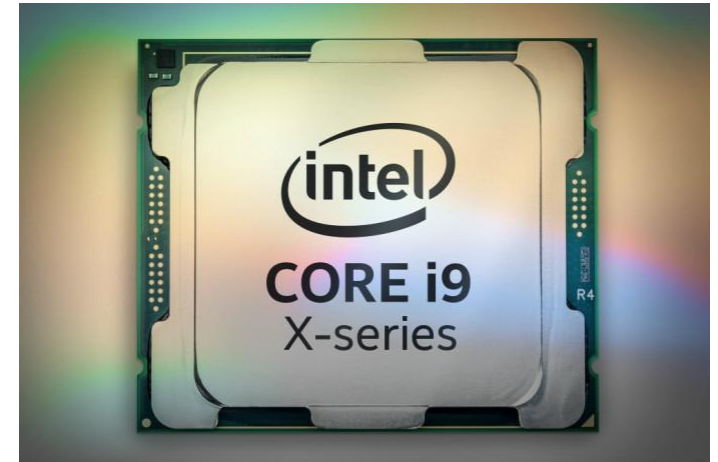
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Aspect	SMP (Symmetric)	AMP (Asymmetric)
Processor Role	All processors equal	One master, several slaves
OS Control	Shared by all CPUs	Controlled by master only
Memory Access	Shared, uniform	Shared or separate
Scheduling	Dynamic	Static
Fault Tolerance	High	Low
Communication	Shared memory	Shared memory or message passing
Scalability	Moderate	Poor
Common Use	Servers, HPC	Embedded, RTOS, IoT

Real-World SMP Example

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- ❑ **Intel Core i9 / AMD Ryzen:**
 - ❑ Each core acts as an independent CPU.
 - ❑ Shared L3 cache and main memory.
 - ❑ OS distributes threads dynamically across cores.
- ❑ **Operating System Example:**
 - ❑ Windows or Linux uses symmetric scheduling to balance threads.
- ❑ **Result:** Efficient utilization of CPU cores for multitasking and parallel workloads.



Real-World AMP Example

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□ Automotive ECUs:

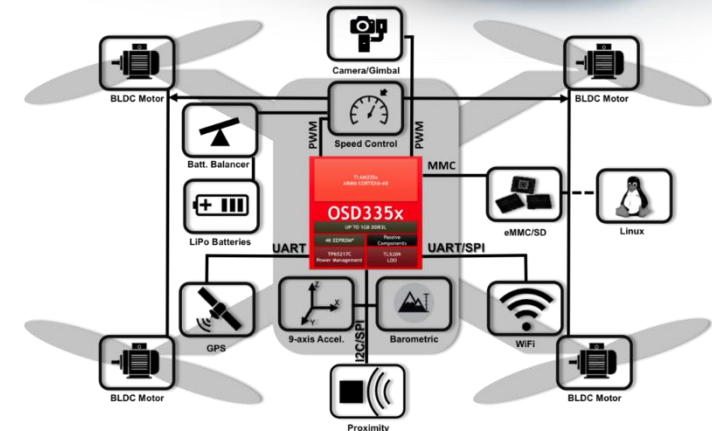
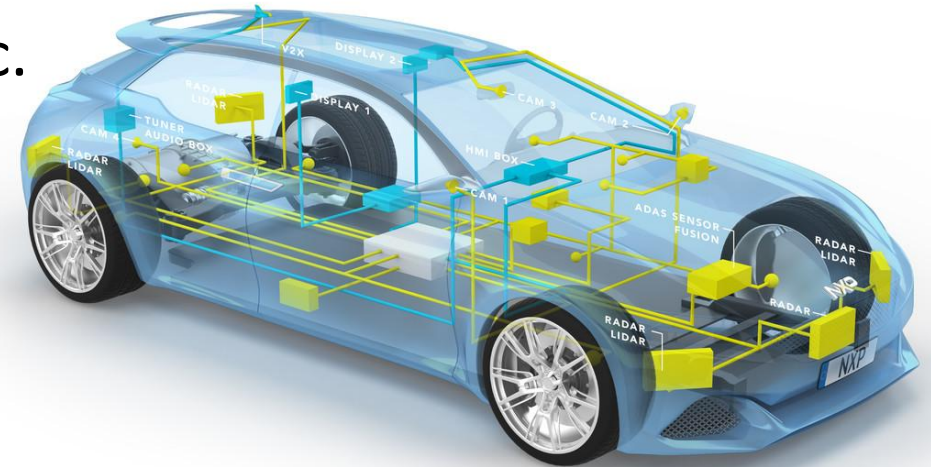
- ▣ Master CPU handles overall control logic.
- ▣ Slave CPUs handle subsystems like fuel injection or braking.

□ Mobile SoCs (ARM big.LITTLE):

- ▣ Big cores for performance tasks.
- ▣ LITTLE cores for power efficiency.

□ Drone Control Systems:

- ▣ Master handles navigation, slaves manage sensor data or motor control.



Summary

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- **Multiprocessing:** Multiple CPUs increase performance and reliability.
- **SMP:**
 - ▣ Equal processors
 - ▣ Shared OS and memory
 - ▣ Dynamic task scheduling
- **AMP:**
 - ▣ Master-slave architecture
 - ▣ Static assignment
 - ▣ Simpler but less flexible
- **Modern systems** blend both approaches for efficiency and power balance. (Hybrid Multiprocessing (HMP))

Thanks

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“One small positive action can
change the course of your
entire life.”

